Tiny Cracks Help Solve a Mystery on Asteroid Bennu

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In 2018, NASA's OSIRIS-REx **spacecraft** reached an **asteroid** called Bennu. Scientists were surprised. Instead of smooth, sandy ground, Bennu was covered in huge **boulders**.

That was a puzzle. Earlier measurements showed Bennu's surface heats up and cools down fast, the way beach sand does. Big rocks usually hold heat and cool slowly, like concrete.

The **spacecraft** brought rock **samples** back to Earth. A team led by Andrew Ryan of the University of Arizona studied them. They found the rocks were full of tiny empty spaces and networks of cracks.

Scientists in Japan heated small spots on the **samples** with a laser to see how heat spread. At NASA's Johnson Space Center in Houston, scientists sealed the rocks in **airtight** containers and used 3D X-ray scans to see inside without harming them. The results show that the size and inside structure of the rocks matter a lot. The study was published in the journal Nature Communications.

Word Treasure

asteroid -- A big space rock that travels around the sun.
spacecraft -- A vehicle that travels in space.
boulders -- Very large rocks.
samples -- Small pieces collected for scientists to study.
airtight -- Sealed so no air can get in or out.

Background you need

Asteroids are space rocks left over from when the planets formed.

NASA sent a spacecraft to the asteroid Bennu to collect rocks and bring them back to Earth.

Scientists can learn what a surface is like by seeing how fast it warms and cools.

Break it down

WHO: NASA scientists and a team led by Andrew Ryan
WHAT: Rocks from asteroid Bennu are full of tiny cracks and empty spaces
WHERE: Asteroid Bennu, and labs in the U.S. and Japan
WHY: To learn why Bennu's rocky surface cools down so fast

Why it matters

Studying space rocks helps us learn how our solar system was made.

Test what you remember

1. What was Bennu covered in?

- (A) Huge boulders
- (B) Water
- (C) Snow
- (D) Trees

2. Which spacecraft reached Bennu in 2018?

- (A) Voyager
- (B) OSIRIS-REx
- (C) Hubble
- (D) Apollo 11

3. How do big rocks usually behave with heat?

- (A) They cool very fast
- (B) They glow
- (C) They melt right away
- (D) They hold heat and cool slowly

4. What did scientists find inside the rocks?

- (A) Ice cream
- (B) Gold
- (C) Fossils
- (D) Tiny empty spaces and cracks

5. What did scientists in Japan use to heat small spots?

- (A) An oven
- (B) The sun
- (C) A candle
- (D) A laser

6. Why were the rocks sealed in airtight containers?

- (A) To make them bigger
- (B) To paint them
- (C) To keep them clean and unharmed
- (D) To make them float

Answer key (try the quiz first!): 1: A 2: B 3: D 4: D 5: D 6: C

Questions to think about

MY THOUGHTS (at least 20 words):